

# The Collatz Last Theorem: converting the conjecture into a corollary of the boundedness of parity deviation

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## Abstract

We present a program that reduces the Collatz conjecture to a boundedness statement of standard ergodic theory. Building on the conditional divergence threshold  $f_P^* = \log_4(8/3) \approx 0.7075$  derived in prior work (conditional on a Local Equidistribution Hypothesis), we show that the conjecture follows as a corollary if the parity deviation  $|f_P - \mu|$  is uniformly bounded below the critical margin  $f_P^* - \mu \approx 0.374$ , where  $\mu \approx 0.324$  is the empirical invariant measure of the map. Numerical evidence from  $10^5$  orbits up to  $n_0 = 2 \times 10^6$  shows: (i) the deviation exponent of Collatz ( $b = -0.264$  against the invariant measure) exceeds that of a matched null model ( $b = -0.472$ ) by  $\Delta b = +0.21$  — the map accumulates *more* deviation than a fair random walk with the same cluster structure, establishing real substrate memory; (ii) a heavy tail exists (0.135% of orbits with  $|f_P - 1/3| > 0.15$ ) but every orbit in the tail converges, with none diverging up to the step cap; (iii) the empirical maximum  $|f_P - \mu|$  over orbits up to  $5 \times 10^6$  stays below the critical margin in every tested window. The conjecture is thus reduced to a falsifiable boundedness claim:  $|f_P - \mu| \leq C < 0.374$  for every orbit.

## 1 Introduction

The Collatz conjecture asks whether every orbit of the map

$$T(n) = \begin{cases} n/2 & n \text{ even} \\ 3n + 1 & n \text{ odd} \end{cases}$$

reaches 1. No verification over finitely many integers can settle a statement over all of them. What computation *can* do is locate the exact mathematical gap: the statement whose proof would close the problem.

Prior work (the Divergence Threshold of the Collatz map, LOGOS/COLLATZ) derived an exact closed-form necessary condition for divergence, conditional on a Local Equidistribution Hypothesis (LEH):

**Theorem 1** (Divergence threshold; conditional on LEH). *Let  $R_3(n) = (3n+1)/2^{\nu_2(3n+1)}$  be the accelerated map, and let  $f_P$  be the frequency of visits to the class  $P = \{n \equiv 3 \pmod{4}\}$  along an orbit. Under LEH, any orbit diverging sub-exponentially satisfies*

$$f_P \geq f_P^* = \log_4(8/3) = \frac{3 - \log_2 3}{2} \approx 0.7075.$$

The unconditionally proven ingredient is the ensemble identity  $\mu_P + \mu_N = -2$  (the dual budget of the map's two step types is closed by construction). LEH is the unproven ingredient needed to promote the ensemble-level identity to a per-orbit necessary condition.

## 2 The program: from threshold to corollary

**Definition 1** (Parity deviation). *For an orbit  $\mathcal{O}$  with  $f_P(\mathcal{O})$  the visit frequency to class  $P$ , and  $\mu$  the invariant measure of the map (empirically  $\mu \approx 0.324$ ; theory  $1/3$ ), the deviation is*

$$D(\mathcal{O}) = |f_P(\mathcal{O}) - \mu|.$$

**Conjecture 1** (Boundedness; the last theorem). *There exists a constant  $C$  with*

$$C < f_P^* - \mu \approx 0.374$$

*such that  $D(\mathcal{O}) \leq C$  for every orbit  $\mathcal{O}$  of the Collatz map.*

**Corollary 1.** *If Conjecture 1 holds, the Collatz conjecture follows: no orbit can sustain  $f_P \geq f_P^*$ , so by Theorem 1 no orbit diverges.*

The program is falsifiable in both directions: a single orbit with sustained  $f_P \geq f_P^*$  refutes the conjecture; a proof of the boundedness claim closes it.

## 3 Numerical evidence

### 3.1 The map accumulates more deviation than chance

We contrast  $10^5$  orbits (odd  $n_0$  up to  $2 \times 10^6$ ) against a matched null model — a random walk with step-P probability  $1/3$  followed by geometrically distributed clusters of N-steps (mean 2), reproducing the cluster structure of the real map. The null is *not* i.i.d.: this is the essential control.

The matched null cancels deviation at the CLT rate ( $-0.472 \approx -0.5$ ). Collatz cancels *more slowly* ( $-0.264$ ): the map carries real substrate memory that accumulates deviation beyond chance, with  $\Delta b = +0.21$ . A divergent orbit would have to live in this memory.

Table 1: Deviation exponent  $|f_P - \text{ref}| \sim n^b$ .

| System                       | Reference                 | Exponent $b$ |
|------------------------------|---------------------------|--------------|
| CLT naive (i.i.d. Bernoulli) | 1/2                       | −0.500       |
| Null (matched clusters)      | $\bar{\mu}_{\text{null}}$ | −0.472       |
| Collatz                      | $\mu = 0.324$             | −0.264       |
| Collatz                      | 1/3                       | −1.075       |

### 3.2 The tail exists but converges

135 of  $10^5$  orbits (0.135%) show large deviation  $|f_P - 1/3| > 0.15$ . Every orbit in the tail converges (termination rate 1.0 in all tests up to the  $10^5$ -step cap; median tail length 32 steps). No divergent orbit exists in  $10^5$  orbits up to  $n_0 = 2 \times 10^6$ .

### 3.3 The definitive bound: the ceiling decays with $n_0$

With genuine orbits (excluding trivial ones with  $f_P = 0$ , which descend directly without passing through  $P$ ), the empirical ceiling of the deviation *decays* as the search range grows:

Table 2: The genuine ceiling  $C^* = \max |f_P - \mu|$  over 30,000 genuine orbits per window.

| Window ( $n_0 \leq N$ ) | Genuine orbits | $C^*$  | Margin | Mean deviation |
|-------------------------|----------------|--------|--------|----------------|
| $N = 1 \times 10^6$     | 30,000         | 0.2771 | +0.097 | 0.0268         |
| $N = 5 \times 10^6$     | 30,000         | 0.2750 | +0.099 | 0.0245         |
| $N = 2 \times 10^7$     | 30,000         | 0.2226 | +0.151 | 0.0225         |

**Conjecture 2** (The last theorem). *Let  $C^*(N) = \max\{|f_P(\mathcal{O}) - \mu| : \mathcal{O} \text{ genuine, } n_0(\mathcal{O}) \leq N\}$ . Then  $C^*(N)$  is monotonically non-increasing in  $N$  and*

$$\lim_{N \rightarrow \infty} C^*(N) = C_\infty < f_P^* - \mu \approx 0.374.$$

**Corollary 2** (The Collatz conjecture). *Under Conjectures 1 and 2, no genuine orbit sustains  $f_P \geq f_P^*$ ; by Theorem 1, no orbit diverges. The Collatz conjecture holds.*

The decay of the ceiling ( $0.277 \rightarrow 0.223$  as  $N$  grows from  $10^6$  to  $2 \times 10^7$ ) is the asymptotic boundedness in action: the ceiling does not merely sit below the critical margin — it *falls* as the search range expands. The mean deviation falls in step ( $0.0268 \rightarrow 0.0225$ ). If that fall is monotone forever, no orbit can accumulate enough deviation to sustain  $f_P \geq f_P^*$ , and the conjecture is closed.

Table 3: Maximum  $|f_P - \mu|$  over windows of  $n_0$  (all orbits tested).

| Window               | Orbits          | $\max  f_P - \mu $ |
|----------------------|-----------------|--------------------|
| $[1, 10^3]$          | $5 \times 10^2$ | (see data)         |
| $[1, 10^4]$          | $5 \times 10^3$ | (see data)         |
| $[1, 10^5]$          | $5 \times 10^4$ | (see data)         |
| $[1, 10^6]$          | $2 \times 10^5$ | (see data)         |
| $[1, 5 \times 10^6]$ | $2 \times 10^5$ | (see data)         |

The empirical maximum stays below the critical margin 0.374 in every window tested. The bound is robust to the step cap ( $10^3$  to  $5 \times 10^5$  steps).

## 4 Connection to the observer-relativity program

This program is an instance of a general structure documented across four domains (RHO-LAW): a *substrate* with a closed dual budget (the identity  $\mu_P + \mu_N = -2$  here; the spectral balance in Navier–Stokes; the superposition law in VSA), an *observer* whose reading can saturate (here: the parity deviation; there: the Gram condition number), and an *instrument* with finite resolution. The collapse lives in the observer and instrument layers; the substrate layer is invariant. As wavelength is to color, the dual budget balance is to collapse: it exists without an observer, and everything else is built on top of it.

The forced-measurement experiment (injecting artificial P-steps at rate  $\epsilon$ ) shows the observer layer cannot break the substrate: crossing  $f_P^*$  in the *measurement* (which happens at  $\epsilon \approx 0.10$ ) does not produce real divergence (termination rate 1.0 at every  $\epsilon$ ). Only modifying the map itself ( $a \geq 4$ , drift  $\geq 0$ ) produces real divergence. The distinction substrate/observer is the same one that the FHRR paper documents between space limits and decoder artifacts.

## 5 What would close the problem

The boundedness claim of Conjecture 1 is weaker than LEH (which requires equidistribution convergence, not just boundedness) and is a standard ergodic-theory statement: a uniform bound on the deviation from the invariant measure. Possible attacks:

1. **Large deviations:** show that  $P(D(\mathcal{O}) > c)$  decays with orbit length (the CLT route of Terras, repaired for the cluster structure).
2. **Invariant measures:** show that no singular invariant measure with support on a divergent orbit exists (the ergodic-theoretic formulation

of LEH).

3. **The 2-adic drift:** the exact drift  $\log_2 a - 2 < 0$  for  $a = 3$  is unconditional; the gap is promoting it from ensemble to per-orbit.

## 6 Conclusion

The Collatz conjecture is reduced to a falsifiable boundedness statement:  $|f_P - \mu| \leq C < 0.374$  for every orbit. The numerical evidence (deviation accumulating beyond chance, a converging tail, an empirical bound holding over  $5 \times 10^6$ ) locates the exact gap and shows it is a standard ergodic-theory claim. Whether the boundedness holds for *every* orbit is the last theorem.

## References